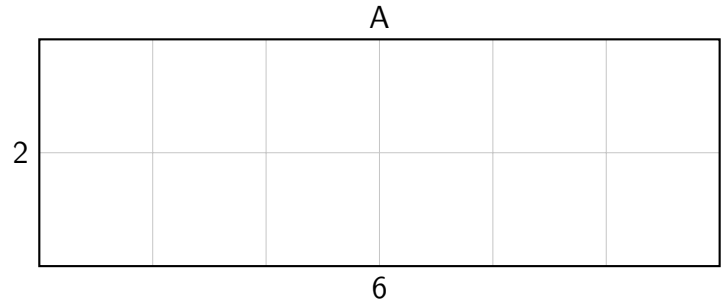
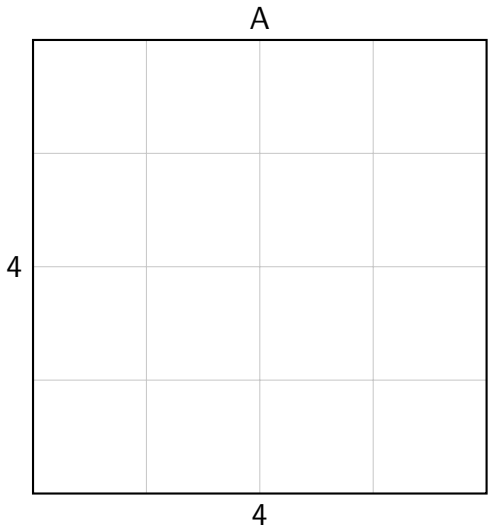
Each grid-square edge is 1 unit. Rectangles turned around count as the same rectangle.

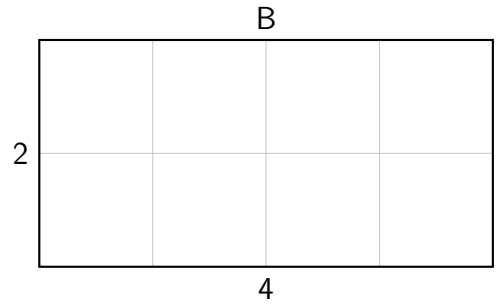
Problem 1: Use 24 square tiles to make every different rectangle you can. Draw your collection. Which rectangle comes closest to being a square?

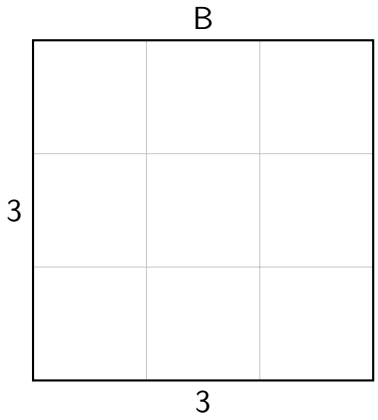


Problem 2: Cut each rectangle into pieces and fit all its pieces inside the matching square, without overlaps. Can you leave exactly one square uncovered in each case? Find the side length of each uncovered square.

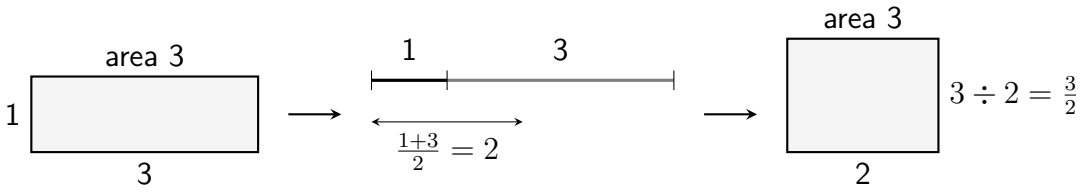








A reshaping move keeps the area: average the two sides, then divide the area by that average to get the other new side. We are making the rectangle closer to a square and trying to trap the square's side length.

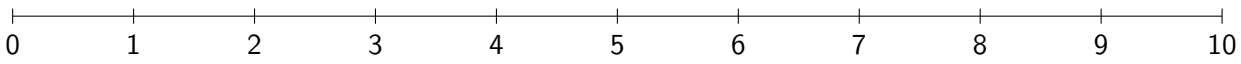


Keep side lengths exact: $20 \div 6 = \frac{10}{3}$. Use fractions as labels; a calculator estimate can help place them. Check any proposed decimal bound by squaring it.

Problem 3: Reshape each rectangle once. Mark the old and new side lengths on its number line. Which two of those four lengths give the tightest guaranteed range for the side of a square with that area?

2 by 6 area 12

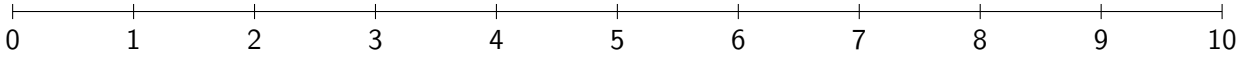
New sides: _____ and _____



Square side between _____ and _____

3 by 6 area 18

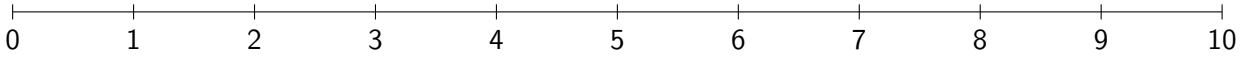
New sides: _____ and _____



Square side between _____ and _____

2 by 10 area 20

New sides: _____ and _____



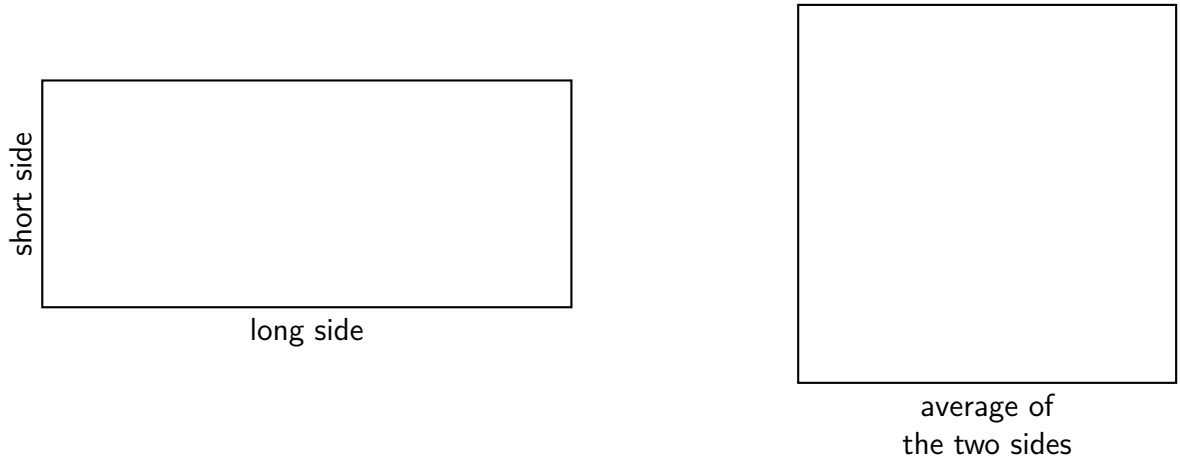
Square side between _____ and _____

Problem 4: Cut out the four cards from a spare copy of this page. They show four stages of reshaping one rectangle, with exact side lengths. Put them in order so each card follows from one reshaping move. Which card gives the tightest guaranteed range for the side of a square of area 12? Show why that square's side cannot lie outside your range.

A $\frac{24}{7}$ by $\frac{7}{2}$	B 2 by 6
C $\frac{336}{97}$ by $\frac{97}{28}$	D 3 by 4

Problem 5: Find two decimals one hundredth apart that trap the side of the square of area 12. Could either decimal be the exact side length? Give exact area checks for your answer.

Problem 6: For any rectangle, is the square whose side is the average of the rectangle's sides always at least as large in area? Explain with a cut-and-rearrange picture. How much extra area does the square have?



Diagrams not to scale.

Problem 7: Start with an unequal-sided rectangle and keep reshaping it. Can the guaranteed range for the equal-area square's side become as narrow as you want? Can a move ever produce the exact square? Explain why your answers must hold for every unequal-sided rectangle.